# Public Data Download Guide

Ship Partners Ltd. · October 2026

- **Project:** Real-time Testbed and Data on Real Ship Connected via Starlink
- **Period:** September 2024 – August 2028
- **Host:** Seoul National University
- **Sponsor:** Lloyd's Register Foundation (LRF)
- **Vessel and data:** Helikon Shipping
- **Scope:** navigation data of an in-service bulk carrier — navigation sensors, ship motion, radar, cameras, wave radar; hull form and mass distribution
- **Collection:** 2 years, September 2025 – August 2027
- **Purpose:** research and development only
- **This guide:** how to download the data; the sample package

- **Sample:** 2025-11-15_heavy_sea, 1.2 GB — DTI player + data, 15 Nov 2025 00:49:00–00:51:20 UTC
- **Full data:** shippartners.synology.me (HTTP, FTP, SFTP)
- **Website:** https://github.com/JunghyunKimKR/realship-data
- **Account, ports:** by e-mail — request to kim@shippartners.jp

## 1. Overview

### 1.1 Vessel

| Item | Value |
|---|---|
| Vessel | MV AREOPAGUS, IMO 9985667, Barbados, ClassNK |
| Type | Ultramax bulk carrier, 5 holds, DWT 66,648 t, GT 36,560 |
| Dimensions | LOA 199.99 m, Lpp 196.1 m, B 32.25 m, D 19.15 m, summer draught 13.83 m |
| Delivered | 15 Aug 2025 |
| Main engine | MAN B&W 6S50ME-C9.7-EGRBP; MCR 6,990 kW × 94.8 rpm, NCR 5,940 kW × 89.8 rpm |
| Design speed | 14.0 kn |
| Propeller, generators | FPP, D 6.6 m; 3 × 560 kWe |

### 1.2 Measurement Systems

| System | Equipment | Recorded as | Rate |
|---|---|---|---|
| Radar | Furuno FAR-2338S-NXT (S), FAR-2328-NXT (X); 4 echo channels | Radar/*.bin | ~0.4 rev/s |
| Cameras | 5 × Arges GC-F12BN fisheye 12 MP (4 bridge wings, 1 foremast) | Camera/*.mkv, H.265 4000 × 3000 | ~10 fps |
| IMU | SBG Ellipse-A; 10.13 m fwd of A.P., 2.45 m to port, 34.12 m above BL | $PSBGI, $PASHR | 10 Hz |
| Navigation | GNSS × 2, gyro, speed log, anemometer, echo sounder, AIS | NMEA | 1–10 Hz |
| Wave radar | Furuno WV-100ST (on the S-band radar) | RPT/WaveAnalyzer, $PFEC,RAwv2 | ~80 s |

All recorded by the DTI (Digital Twin Instance) on the onboard server. All times UTC.

## 2. Data Holdings

- **Collection:** 2 years, Aug 2025 – Aug 2027 (ongoing); shore copy up to 22 May 2026 (status October 2026)
- **Transfer:** onboard NAS disks carried ashore (too large for the satellite link)

| Data | Folder · file | Period (UTC) | Days | Volume |
|---|---|---|---|---|
| NMEA log (navigation, motion, AIS, wave radar) | Logs · Log_Daily_yyyyMMdd.zip | 21 Aug 2025 – 22 May 2026 | 274 | 23.2 GB zip (150 GB text) |
| Radar echo, 4 channels | Radar · Radar_Daily_yyyyMMdd_HHmmss.bin | 14 Nov 2025 – 22 May 2026 | 148 | 7.8 TB (~50 GB/day) |
| Camera video, 5 cameras | Camera · yyyy-MM-dd/yyyy-MM-dd_HH-mm.mkv (30 min) | 18 Aug 2025 – 17 May 2026 | 189 (2,980 h) | 50.6 TB (~20 GB/h) |
| Wave radar results, 2-D spectra | RPT/WaveAnalyzer · result/, spectrum/ | 14 Aug 2025 – 11 May 2026 | 204 | 1.85 GB |
| Mass distribution | RPT/LoadingConditions · MassDistribution-<condition>.xlsx | light ship, Voy01–Voy07 and voyage legs (<condition>-Leg<n>) | — | 0.9 MB |
| Hull form | ABS · .stl | — | — | 1.0 MB |

NMEA log, radar and camera on the same day: 100 days (14 Nov 2025 – 11 May 2026).

Days with data per month (a day counts when any data of that type exists):

| Month | NMEA log | Radar | Camera (days / hours) | Wave radar |
|---|---|---|---|---|
| Aug 2025 | 11 | — | 14 / 320 h | 9 |
| Sep 2025 | 30 | — | 7 / 142 h | 12 |
| Oct 2025 | 30 | — | 23 / 407 h | 17 |
| Nov 2025 | 30 | 10 | 23 / 258 h | 27 |
| Dec 2025 | 31 | 17 | 31 / 568 h | 23 |
| Jan 2026 | 31 | 22 | 18 / 288 h | 22 |
| Feb 2026 | 28 | 25 | 20 / 238 h | 25 |
| Mar 2026 | 31 | 28 | 20 / 272 h | 28 |
| Apr 2026 | 30 | 30 | 16 / 198 h | 30 |
| May 2026 | 22 | 16 | 17 / 289 h | 11 |
| **Total** | **274** | **148** | **189 / 2,980 h** | **204** |

- **NMEA log:** continuous from 21 Aug 2025, missing 12 Oct 2025
- **Old log format:** up to 9 Oct 2025 the time prefix has 1-s resolution (`2025-09-14T23:59:59, $…`); DTI playback from 10 Oct 2025
- **Radar:** from 14 Nov 2025; a file's span can exceed its echo span (status sentences continue after the radar is off)
- **Camera:** 10–24 h per recorded day (average 16 h), 30-min files

## 3. Downloading the Full Data

- **Server:** shippartners.synology.me
- **Protocols:** HTTP (browser, single files); FTP / SFTP for bulk, e.g. FileZilla
- **Account, port:** by e-mail; request to kim@shippartners.jp

| Folder | Contents | Download |
|---|---|---|
| ABS | hull form (STL) | first |
| RPT | mass distribution, wave radar results and spectra | first |
| Program | DTI program (reading and replaying the data) | first |
| Logs | NMEA, one zip per day, 60–100 MB | dates you need |
| Camera | video, one folder per day, ~11 GB per 30-min file | when needed |
| Radar | echo, one file per day, 25–80 GB | when needed |
| SampleCodes | Python examples (streams, radar echo) | when needed |

- One period = the day's log zip + the day's radar file + the 30-min camera files covering it.
- Wave radar files are named by their first record: result file 1–2 days, spectrum file ~2.5 h.

# 4. Sample Dataset

## 4.1 Package Contents

| Path | Size | Contents |
|---|---|---|
| player/ | 179 MB | DTI player (playback only), dtisettings.xml, LICENSE.txt, THIRD-PARTY-NOTICES.txt |
| data/Camera/2025-11-15/2025-11-15_00-49.mkv | 882 MB | 5 video tracks + 2 NMEA subtitle tracks, 00:49:00–00:51:20 |
| data/Radar/Radar_Daily_20251115_004850.bin | 134 MB | 4 channels, 00:48:50–00:51:25 |
| data/Logs/Log_Daily_20251115.zip | 0.3 MB | 25,605 NMEA lines, 00:46:50–00:51:25 (2 min lead-in) |
| data/RPT/WaveAnalyzer/result/…_wave_result.csv | 1.0 MB | wave radar results (original file, 14–16 Nov) |
| data/RPT/WaveAnalyzer/spectrum/…_spectrum.csv | 1.0 MB | 2-D wave spectra (original file, 14–15 Nov) |
| data/RPT/LoadingConditions/MassDistribution-Voy02-Leg2.xlsx | 35 kB | mass distribution of the sample window: Voy02 leg 2 (5.5) |
| data/ABS/bulkcarrier_surface_rc62.stl | 1.0 MB | hull form (approximate, 5.6) |
| data/SampleCodes/ | 17 kB | radar_parser.py, json_viewer.py |
| README.txt, this guide | | |

- Total 1.21 GB. data/ has the same layout as the server.
- Nothing re-encoded: video stream copy, radar blocks and log lines byte copies.

## 4.2 Sample Window

| Item | Value |
|---|---|
| Time | 15 Nov 2025, 00:49:00–00:51:20 UTC |
| Position | 31.14° N 130.27° W, open North Pacific |
| Course, speed | heading 112°, SOG 11.4 kn, STW 14.1 kn |
| Wind | true 8–10 m/s from ~340° (port quarter) |
| Waves (wave radar) | Hs 3.67–3.87 m, T01 ~10 s, Tp 12.5 s, from ~329° |
| Roll | SD 2.48°, max 5.3° |
| Pitch, heave | SD 0.68°, 1.03 m (heave at the IMU) |
| Loading | Voy02 leg 2: laden, 61,026 t |
| Light, traffic | late afternoon, horizon on all cameras; no AIS targets in range |

Why this window: the largest roll in the only camera hour of the day (00:00–01:00 UTC).

# 5. Data Formats

Folder layout (same on the server and in the sample's data/):

```
<data path>\
  Logs\Log_Daily_yyyyMMdd.zip                one .log inside, same name
  Radar\Radar_Daily_yyyyMMdd_HHmmss.bin      file start time in the name
  Camera\yyyy-MM-dd\yyyy-MM-dd_HH-mm.mkv     file start (recorder clock) in the name
  RPT\WaveAnalyzer\result\yyyyMMdd\yyyyMMdd_HHmmss_wave_result.csv
  RPT\WaveAnalyzer\spectrum\yyyyMMdd\yyyyMMdd_HHmmss_spectrum.csv
  RPT\LoadingConditions\MassDistribution-<condition>.xlsx
  ABS\<hull>.stl
```

## 5.1 NMEA Log

- **Line:** `yyyy-MM-dd HH:mm:ss.fff, <sentence>` = DTI receive time (UTC) + the sentence as received; sentence from character 25

```
2025-11-15 00:49:00.034, $INGLL,3108.2956,N,13016.1715,W,004900.08,A,A*6B
2025-11-15 00:49:00.093, $PSBGI,174642.100,-0.657,0.476,0.155,0.117,0.096,-10.047,*68
2025-11-15 00:49:00.097, $PASHR,,,T,-000.95,+000.06,+000.823,0.322,0.321,1.271,0,0*08
2025-11-15 00:49:00.116, $HEHDT,113.6,T*2A
```

| Sentence | Content | Rate |
|---|---|---|
| $PASHR | IMU attitude: roll, pitch, heave at the IMU | 10 Hz |
| $PSBGI | IMU rates and accelerations (table below) | 10 Hz |
| $INGLL, $INRMC, $INVTG, $INVBW, $INROT, $INTHS | integrated navigation: position, COG/SOG, STW, rate of turn, heading | 2–6 Hz |
| $GPGGA, $GPRMC, $GPVTG, $GPZDA | GNSS — two receivers share talker GP; use $INGLL for position | 1 Hz each |
| $HEHDT, $HEROT | gyro heading, rate of turn (+ = to starboard) | 1 Hz |
| $VDVBW, $VDVLW | speed log: STW, distance | 1 Hz |
| $WIMWV | relative wind, m/s | 1 Hz |
| $SDDPT | depth (none in deep water) | 1 Hz |
| $PFEC,RAwv2 | wave radar summary: whole sea, 1st and 2nd system | ~80 s |
| !AIVDM / !AIVDO | AIS, other ships / own ship (no targets in the sample) | on receipt |

### Motion Signs

| DTI name | Field | Unit | + direction |
|---|---|---|---|
| Roll | $PASHR 4 | deg | port side down |
| Pitch | $PASHR 5 | deg | bow down |
| Heave | $PASHR 6 | m | down (at the IMU, not midship) |
| RotX, RotY, RotZ | $PSBGI 2, 3, 4 | deg/s | roll port down, pitch bow down, turn to starboard |
| AccX, AccY, AccZ | $PSBGI 5, 6, 7 | m/s² | aft, port, down (specific force; AccZ ≈ −9.8 at rest) |

- Field n = n-th comma field after the sentence ID (`split(',')[n]`).
- IMU axes aft / port / down (unit installed facing aft). Values as sent; the DTI does not change signs.
- Checked on the sample (gravity, heave–pitch at the IMU lever, rates vs angles, gyro heading) and on 46 steady turns of 9 days (outward heel).

Python:

```
import zipfile, datetime as dt
z = zipfile.ZipFile("Logs/Log_Daily_20251115.zip")
with z.open(z.namelist()[0]) as f:
    for raw in f:
        line = raw.decode("ascii", "replace").rstrip("\r\n")
        t = dt.datetime.strptime(line[:23], "%Y-%m-%d %H:%M:%S.%f")  # UTC
        sentence = line[25:]
        if sentence.startswith("$PASHR"):
            p = sentence.split("*")[0].split(",")
            roll, pitch, heave = float(p[4]), float(p[5]), float(p[6])
```

## 5.2 Radar Echo (.bin)

- **File:** `[int32 length, little-endian][gzip block]` repeated

- **Block:** < 100 bytes unzipped = status sentence (ASCII, e.g. `$PSPLRR` range, `$PSPLRG` gain); otherwise one spoke of 1,053 bytes:

| Offset | Type | Field | Meaning |
|---|---|---|---|
| 0 | int64 | UTC | .NET DateTime ticks (100 ns from 0001-01-01), UTC |
| 8 | uint8 | Id | 0 S-band primary, 1 S-band secondary, 2 X-band primary, 3 X-band secondary |
| 9 | int16 | Angle | 0 … 8191 (8192 = 360°), relative to the bow |
| 11 | int16 | EchoLength | valid range cells (≤ 1024) |
| 13 | double | TotalRangeNM | range of the EchoLength cells (cell = TotalRangeNM / EchoLength) |
| 21 | double | DisplayRangeNM | range setting (sample: 12 NM primary; 0.5 NM S / 1.5 NM X secondary) |
| 29 | uint8[1024] | Echo | 0 … 255 per cell; first EchoLength cells valid |

- **True bearing:** Angle × 360 / 8192 + gyro heading (`Hdg`) at that time
- **Rotation:** ~2.5 s, 1,300–2,500 spokes of 8,192 bearings; fill empty bearings from neighbours
- **Antennas:** S x 16.37 m, z 45.05 m; X x 15.56 m, z 42.90 m; both on the centre line (`<RADAR>` in the settings)
- **UDP:** the DTI streams the same blocks, one per datagram (5.7); reader: `SampleCodes/radar_parser.py`

```
import gzip, struct, datetime as dt
with open("Radar/Radar_Daily_20251115_004850.bin", "rb") as f:
    while (h := f.read(4)):
        block = gzip.decompress(f.read(struct.unpack("<i", h)[0]))
        if len(block) < 100:
            continue                                  # status sentence
        ticks, ch, angle, n, total_nm, disp_nm = struct.unpack_from("<qBhhdd", block, 0)
        t = dt.datetime(1, 1, 1) + dt.timedelta(microseconds=ticks // 10)
        echo = block[29:29 + n]                       # n cells over total_nm
```

## 5.3 Camera Video (.mkv)

- **File:** 30 min, all 5 cameras (sample: 140 s)
- **Tracks 0–4:** H.265, 4000 × 3000, ~10 fps: STBD_FWD, PORT_FWD, PORT_AFT, STBD_AFT, FOREMAST
- **Tracks 5–6:** SSA subtitles = NMEA stream as recorded by the video recorder
- **Lens:** fisheye 180° × 180°; image circle cut at top and bottom of the 4:3 frame
- **Mounting:** position, swivel and tilt per camera in `<Camera>` of the settings (mm from A.P., y + port)
- **Frame time:** file-name time + absolute pts, on the recorder clock
- **Recorder clock:** 0.2–1.2 s off the DTI log clock depending on the day, drift 0.05–0.1 s/h; sample: log = recorder + 1.14 s
- **Alignment:** match `$PSBGI` in the subtitle tracks with the log (each is unique), take the median difference; the DTI does this itself — `/history/camera` returns frames on the log clock (header `X-Frame-Utc`)

```
import av                                     # PyAV
c = av.open("Camera/2025-11-15/2025-11-15_00-49.mkv")
for s in c.streams:
    print(s.index, s.type, s.codec_context.name)   # 0-4 hevc, 5-6 ssa (NMEA)
for frame in c.decode(video=0):                    # track 0 = STBD_FWD
    rec_time = frame.time          # s after the file-name time (recorder clock)
    img = frame.to_ndarray(format="rgb24")         # 3000 x 4000 x 3
    break
```

## 5.4 Wave Radar (RPT/WaveAnalyzer)

### Result files (…_wave_result.csv)

- One row per analysis, ~80 s; `DATE_TIME` = end of a ~240 s analysis.

- `WAVE_HEIGHT` Hs (m) · `DIRECTION_TOTALWAVE` waves from (° true) · `PERIOD_T01_TOTALWAVE`, `PERIOD_T02_TOTALWAVE` (s) · `M0_TOTALWAVE` energy (radar units) · `…_1stWAVE`, `…_2ndWAVE` wave systems · `MEAN_…` running means · ship position, heading, COG, SOG, true wind (m/s).
- Rows with a time only = failed analyses.

**Spectrum files (…_spectrum.csv)**

- Per analysis: header `<yyyy/mm/dd hh:mm:ss>,Period/Direction,0,10,…,350` (36 × 10°, waves from, true), then 26 rows `<f Hz>,<T s>,<36 values>`, 0.04–0.29 Hz.
- Values are uncalibrated radar units: block sum = `M0_TOTALWAVE` / 100 of the same analysis (sample: 1.008).

**Calibration to S(ω, θ), m²·s/rad²**

1. Remove the background: per frequency, the 25th percentile over the 36 directions (image noise; 17 % of the energy in the sample).
2. Scale so that 4 √m0 = `WAVE_HEIGHT`; m0 = Σ S · Δω · Δθ, Δω = 2π · 0.01 rad/s, Δθ = 10° in rad.

- `WAVE_HEIGHT` lags its block by 2–3 analyses. Time-resolved: scale by the block's `M0_TOTALWAVE` × Σ WAVE_HEIGHT² / Σ M0_TOTALWAVE over ±15 min.

## 5.5 Mass Distribution (RPT/LoadingConditions)

- **File:** MassDistribution-Voy02-Leg2.xlsx — Excel, values only (no formulas, no links); sheets TOTAL, Voyage
- **Condition:** Voy02 leg 2, laden — the loading condition of the sample window
- **Content:** the longitudinal mass distribution in 1-m sections — for each section the weight per metre, its centre of gravity (x_g, y_g, z_g) and its rotational inertia about that centre of gravity (I_xx, I_yy, I_zz)
- **Whole ship:** the sections determine the mass distribution and inertia of the whole ship — displacement, centre of gravity, moments of inertia, radii of gyration (also in the file)

| Sheet | Contents |
|---|---|
| TOTAL | per 1-m section: weight per metre [t/m], cumulative weight, centre of gravity x_g / y_g / z_g, inertia about the section centre of gravity I_xx / I_yy / I_zz per metre; whole ship about its centre of gravity (displacement, LCG / VCG / TCG, inertia, radii of gyration); weight curve |
| Voyage | this file: the legs it covers (legs 2–3, UTC from / to); the condition: displacement, deadweight, centre of gravity, draughts, trim, GM, k_xx |

- **Coordinates:** x from A.P., + forward (Lpp 196.1 m, midship 98.05 m); y + starboard; z from base line, + up
- **Voy02 leg 2, whole ship:** Δ 61,026 t · LCG 103.09 m from A.P. · VCG 7.42 m · k_xx 10.18 m (0.316 B) · k_yy 50.35 m (0.257 Lpp) · k_zz 50.87 m

## 5.6 Hull Form (ABS)

- **File:** bulkcarrier_surface_rc62.stl (Hull 2026-10-07 = file date); binary STL, 19,950 triangles, metres
- **Approximate:** fitted to the reference hydrostatic tables (displacement +0.15…+0.34 % at 4–16 m draught); not the ship's lines
- **Use:** hydrostatics and motions; not for resistance or propulsion
- **Extent:** shell up to the deck edge, open at the deck

| Axis | Definition | Extent |
|---|---|---|
| x | from A.P. (x = 0), + forward | −3.39 … 196.60 m |
| y | from the centre line, + port | ±16.15 m |
| z | from the moulded base line, + up; draught = z + 0.027 m (keel plate) | 0 … 24.34 m |

- STL y + port; mass-distribution Excel y + starboard.

```
import numpy as np
b = open("ABS/bulkcarrier_surface_rc62.stl", "rb").read()
n = int.from_bytes(b[80:84], "little")
rec = np.dtype([("n", "<f4", 3), ("v", "<f4", (3, 3)), ("a", "<u2")])
tri = np.frombuffer(b[84:84 + 50 * n], dtype=rec)
v = tri["v"]          # (n, 3, 3) in m: x from A.P., y + port, z from base line
```

## 5.7 Playing the Sample with the DTI

- **Requires:** Windows 10/11 64-bit, .NET Framework 4.8, 16 GB RAM
- **Limit:** one DTI per PC (HTTP port, multicast streams, RTSP server)

1. Copy the package folder to a local disk (any path).
2. Run `player\DTI.exe`: it opens 15 Nov 2025 and loops 00:49:00–00:51:20.
3. CSV Export in the player bar, or the interfaces below.

- **Settings:** `player\dtisettings.xml`; paths relative to this file — `<PlaybackDataPath>..\data</PlaybackDataPath>`
- **Full data:** downloaded folders under `data\` (or another `<PlaybackDataPath>`); delete `<PlaybackStart>` / `<PlaybackLoop>` to choose dates in the player

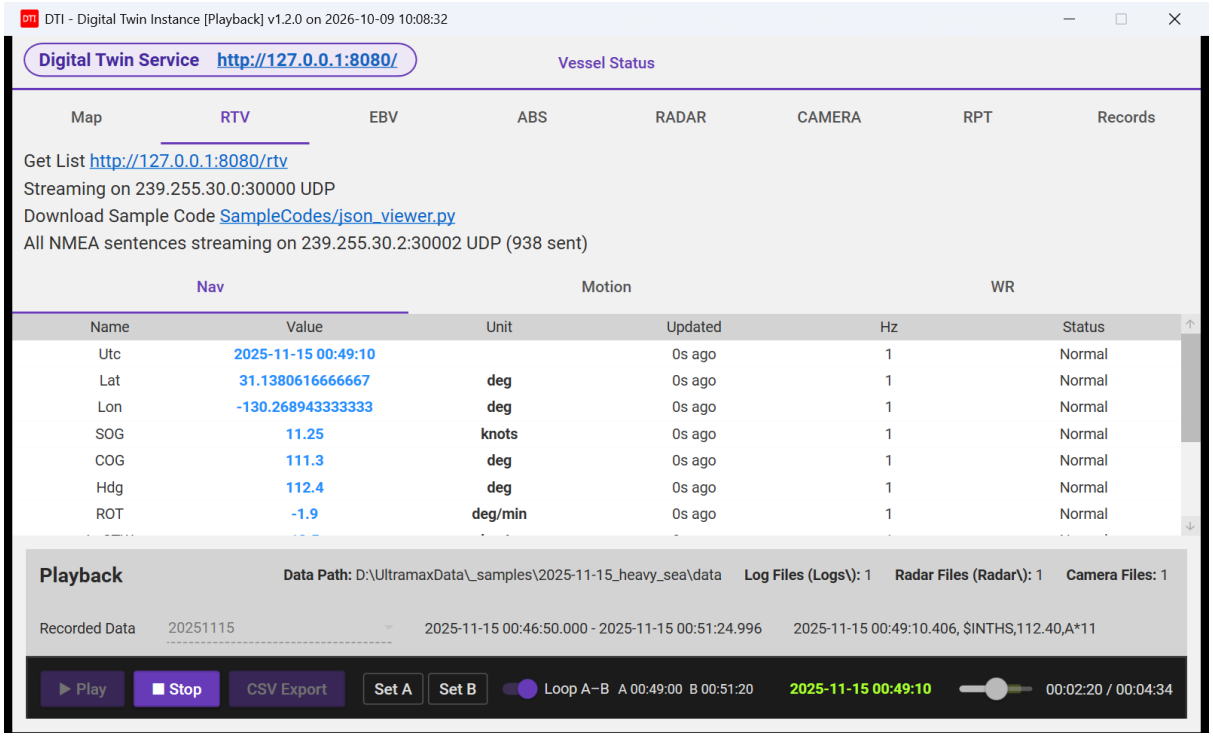



**Figure 1.** DTI replaying the sample: RTV › Nav (live values), player bar with Loop A–B and CSV Export.

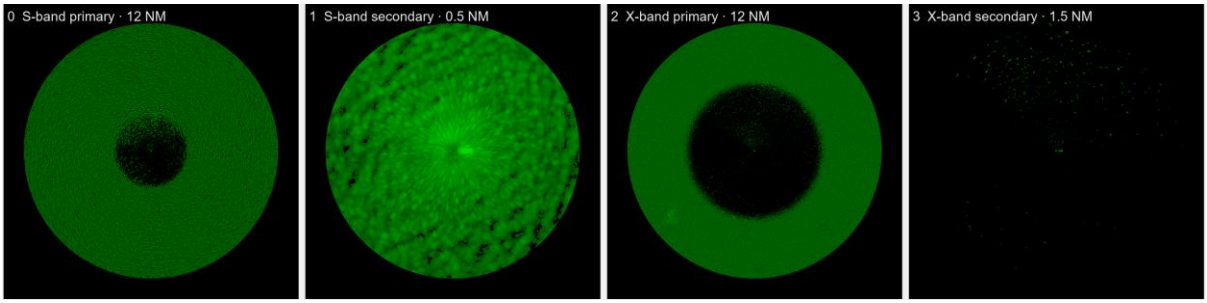



**Figure 2.** DTI › RADAR: the four echo channels, bow up, as they are streamed.

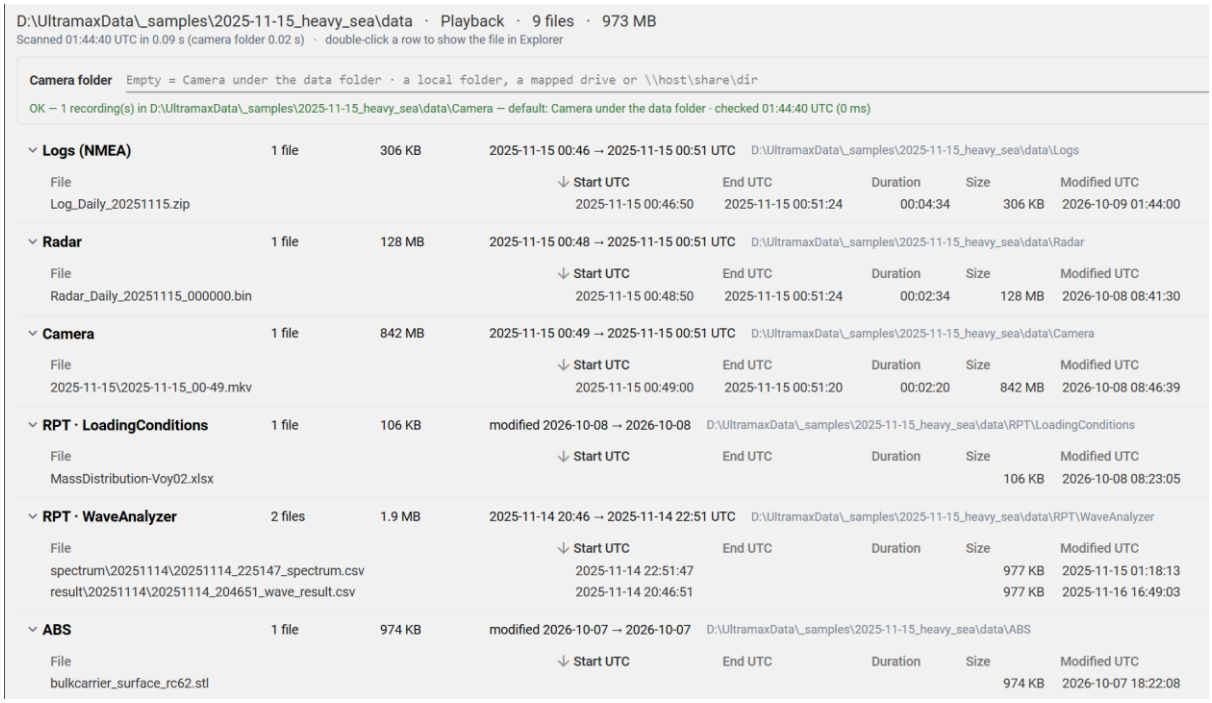



**Figure 3.** DTI › Records: the files of the sample by type, with their UTC spans.

- **Digital Twin Service:** the link at the top left of the DTI (`http://127.0.0.1:8080/`) opens the service page in a browser — every interface of the running DTI on one page
- **All data through it:** live values (`/rtv`, `/ebv`), particulars and model files (`/abs`), report files (`/rpt`), stored history as CSV or JSON (`/history/…`), any file under the data folder (logs, radar, video; HTTP Range), UDP streams, RTSP video

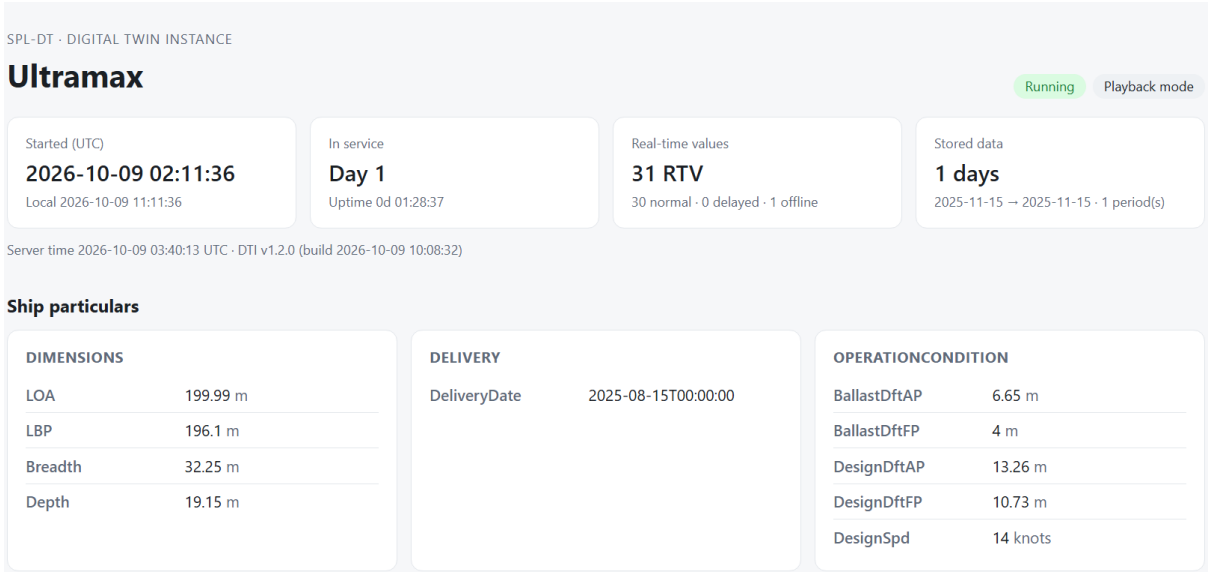
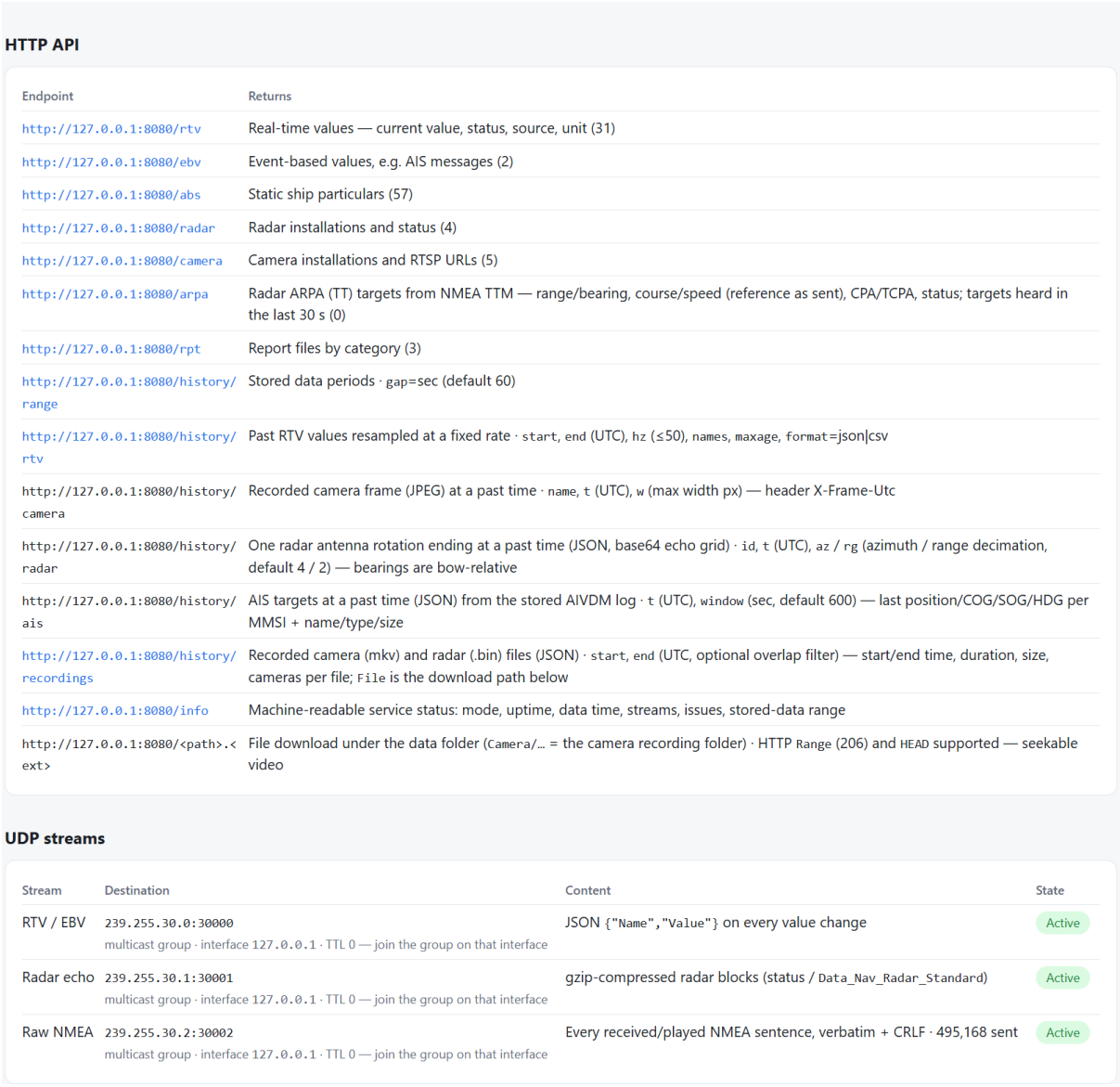



**Figure 4.** Digital Twin Service page, top: service status, stored data period, ship particulars (/abs).



**HTTP API**

| Endpoint | Returns |
|---|---|
| http://127.0.0.1:8080/rtv | Real-time values — current value, status, source, unit (31) |
| http://127.0.0.1:8080/ebv | Event-based values, e.g. AIS messages (2) |
| http://127.0.0.1:8080/abs | Static ship particulars (57) |
| http://127.0.0.1:8080/radar | Radar installations and status (4) |
| http://127.0.0.1:8080/camera | Camera installations and RTSP URLs (5) |
| http://127.0.0.1:8080/arpa | Radar ARPA (TT) targets from NMEA TTM — range/bearing, course/speed (reference as sent), CPA/TCPA, status; targets heard in the last 30 s (0) |
| http://127.0.0.1:8080/rpt | Report files by category (3) |
| http://127.0.0.1:8080/history/range | Stored data periods · gap=sec (default 60) |
| http://127.0.0.1:8080/history/rtv | Past RTV values resampled at a fixed rate · start, end (UTC), hz (≤50), names, maxage, format=json\|csv |
| http://127.0.0.1:8080/history/camera | Recorded camera frame (JPEG) at a past time · name, t (UTC), w (max width px) — header X-Frame-Utc |
| http://127.0.0.1:8080/history/radar | One radar antenna rotation ending at a past time (JSON, base64 echo grid) · id, t (UTC), az / rg (azimuth / range decimation, default 4 / 2) — bearings are bow-relative |
| http://127.0.0.1:8080/history/ais | AIS targets at a past time (JSON) from the stored AIVDM log · t (UTC), window (sec, default 600) — last position/COG/SOG/HDG per MMSI + name/type/size |
| http://127.0.0.1:8080/history/recordings | Recorded camera (mkv) and radar (.bin) files (JSON) · start, end (UTC, optional overlap filter) — start/end time, duration, size, cameras per file; File is the download path below |
| http://127.0.0.1:8080/info | Machine-readable service status: mode, uptime, data time, streams, issues, stored-data range |
| http://127.0.0.1:8080/<path>.<ext> | File download under the data folder (Camera/… = the camera recording folder) · HTTP Range (206) and HEAD supported — seekable video |

**UDP streams**

| Stream | Destination | Content | State |
|---|---|---|---|
| RTV / EBV | 239.255.30.0:30000<br>multicast group · interface 127.0.0.1 · TTL 0 — join the group on that interface | JSON {"Name","Value"} on every value change | Active |
| Radar echo | 239.255.30.1:30001<br>multicast group · interface 127.0.0.1 · TTL 0 — join the group on that interface | gzip-compressed radar blocks (status / Data_Nav_Radar_Standard) | Active |
| Raw NMEA | 239.255.30.2:30002<br>multicast group · interface 127.0.0.1 · TTL 0 — join the group on that interface | Every received/played NMEA sentence, verbatim + CRLF · 495,168 sent | Active |

**Figure 5.** Same page, lower part: every HTTP endpoint with its parameters, and the UDP streams (group, interface, state); the RTSP camera URLs and the radars follow below.

- **CSV Export (player bar; enabled when stopped):** rate ≤ 10 Hz, UTC start / end (several days allowed), the RTVs to include
- **Output:** `Desktop\Export-<start>-<end>.csv` — Timestamp + one column per RTV, one row per 1/Hz s; a value = the last one received, empty when older than 3 rows (e.g. 1-Hz values at 10 Hz)
- **Sample window at 10 Hz, all 31 RTVs:** 1,401 rows, 0.3 MB — the same as `/history/rtv?format=csv`

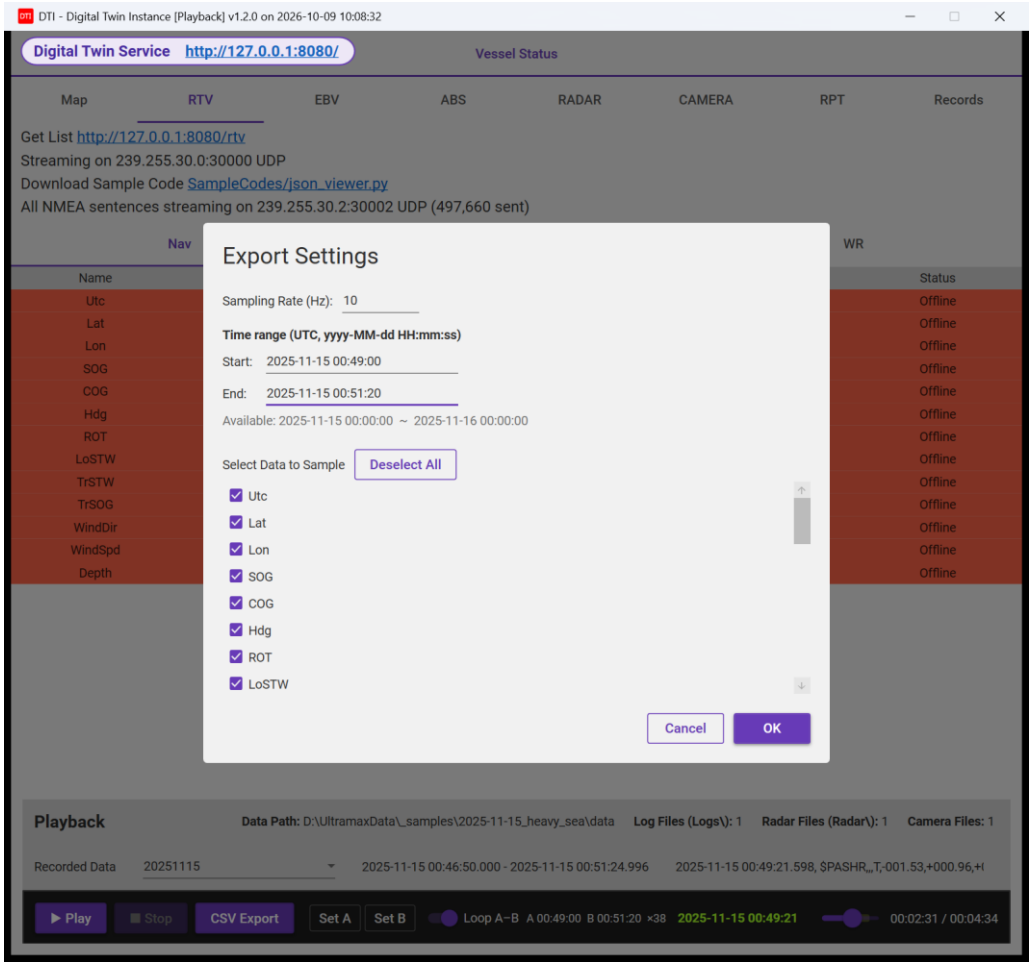



**Figure 6.** CSV Export: sampling rate, UTC range and the RTVs to sample.

Interfaces while the DTI runs (same in playback and on board):

| Interface | Contents |
|---|---|
| HTTP :8080 | `/rtv` values · `/ebv` events (AIS) · `/abs` particulars, model files · `/radar` · `/camera` · `/rpt` report files · `/info` status (JSON) |
| HTTP history | `/history/range` · `/history/rtv?start=&end=&hz=&names=&format=csv` (≤ 50 Hz) · `/history/camera?name=&t=` (JPEG) · `/history/radar?id=&t=` · `/history/ais?t=` · `/history/recordings` — any recorded date |
| UDP 239.255.30.0:30000 | every RTV / EBV change, JSON `{"Name", "Value"}` — `SampleCodes/json_viewer.py` |
| UDP 239.255.30.1:30001 | radar echo, one gzip block per datagram (5.2) — `SampleCodes/radar_parser.py` |
| UDP 239.255.30.2:30002 | every NMEA sentence as received, CR LF |
| RTSP :554 | camera video on the playback clock (MediaMTX); URLs in `/camera` |

- Multicast is sent on this PC only (127.0.0.1, TTL 0): bind the port and join the group on 127.0.0.1. The sample codes do this by default (press Enter at the address prompt).

Example — motion of the sample window at 10 Hz:

```
import urllib.request
url = ("http://127.0.0.1:8080/history/rtv?format=csv&hz=10&names=Roll,Pitch,Heave,Hdg"
       "&start=2025-11-15T00:49:00Z&end=2025-11-15T00:51:20Z")
print(urllib.request.urlopen(url).read().decode()[:160])
# Timestamp,Roll,Pitch,Heave,Hdg
# 2025-11-15T00:49:00.000Z,-0.89,0.02,0.862,113.4
# 2025-11-15T00:49:00.100Z,-0.95,0.06,0.823,113.4
```

# 6. Sample Results

From the package files and the DTI history interface; Figures 7–8 are screens of the AI Officer web application (same ABS file).

## 6.1 Hull Form

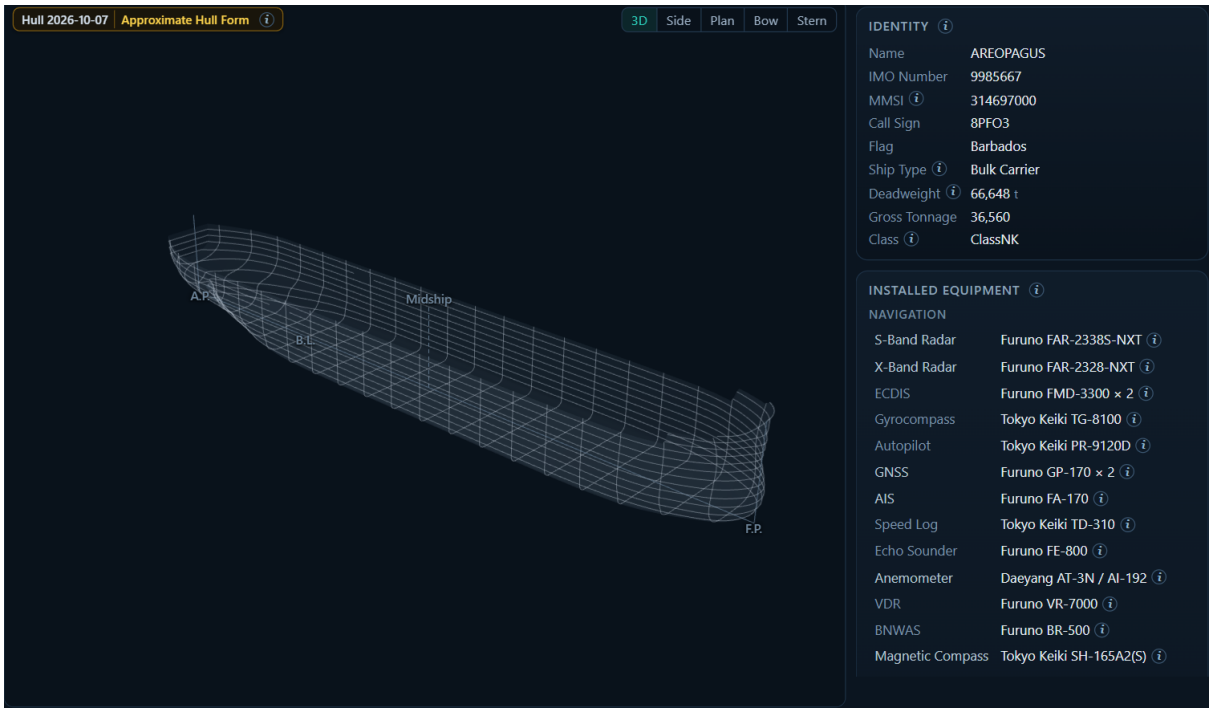



**Figure 7.** Hull form (Hull 2026-10-07, approximate): stations and waterlines; identity and equipment from the DTI (/abs).

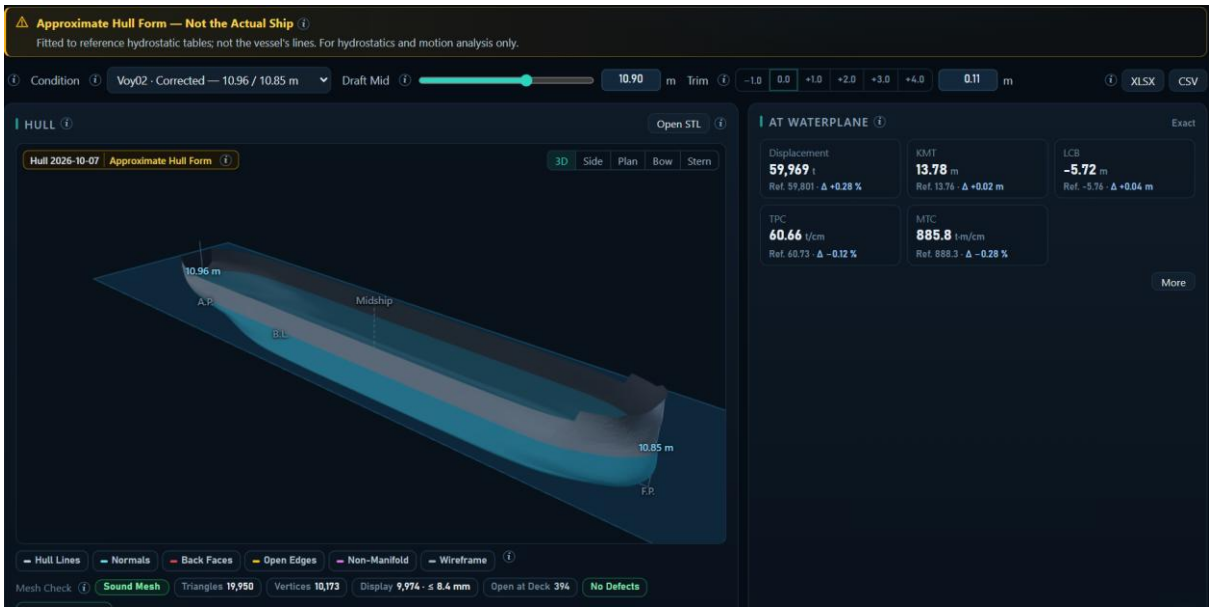



**Figure 8.** Hull STL at the Voy02 departure draughts (10.96 m A, 10.85 m F): 59,969 t, +0.28 % vs the reference hydrostatic table; mesh check clean.

## 6.2 Mass Distribution (Voy02, Leg 2)

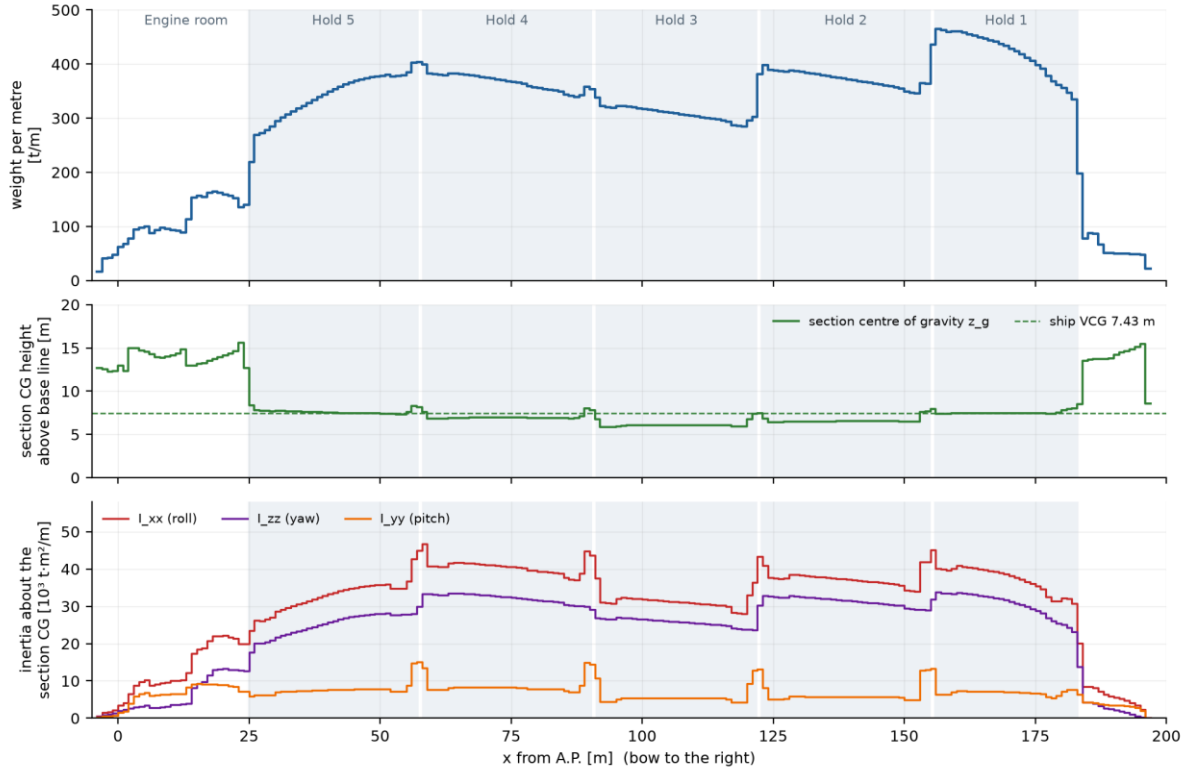



**Figure 9.** MassDistribution-Voy02-Leg2.xlsx, TOTAL: for each 1-m section the weight per metre, the height of its centre of gravity and its rotational inertia about that centre of gravity; together they give the whole ship — Δ 61,026 t, LCG 103.09 m from A.P., VCG 7.42 m, k_xx 10.18 m, k_yy 50.35 m, k_zz 50.87 m.

## 6.3 Cameras




**Figure 10.** Five cameras at 00:50:00 UTC (log clock). Wings look fore and aft on each side; foremast looks aft. Swell from the port quarter.

## 6.4 Radar

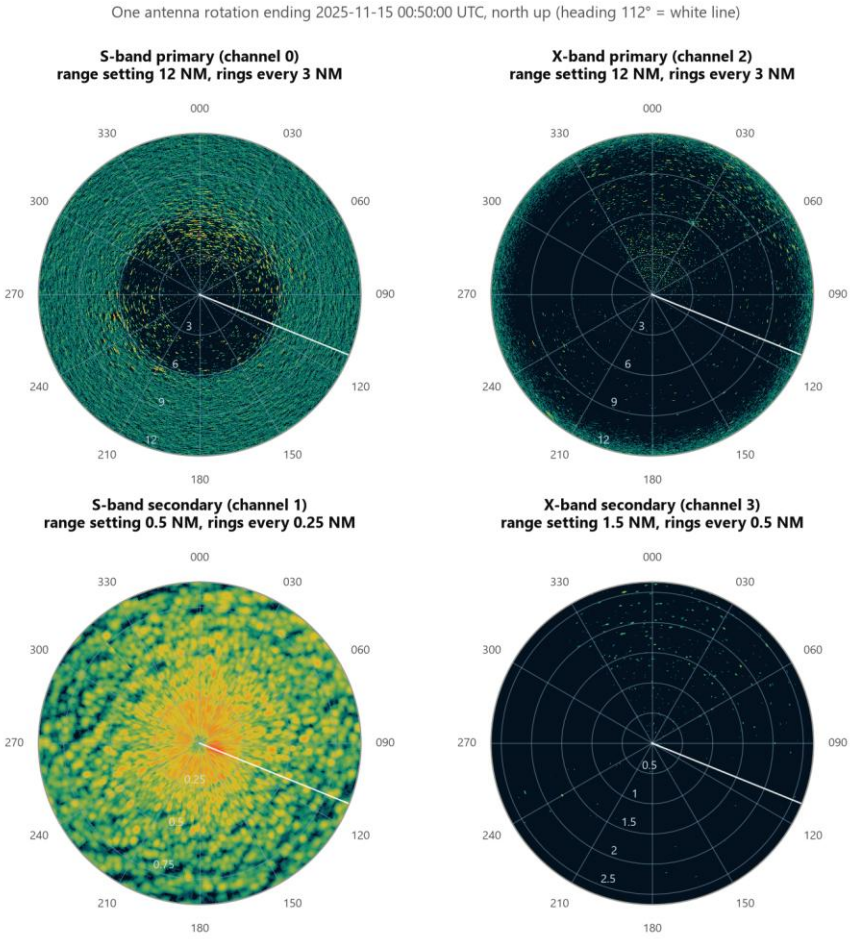



**Figure 11.** One rotation per channel ending 00:50:00 UTC, north up (rotated by Hdg). S-band secondary (0.5 NM) = wave radar input; crests from the NW. No ships in range.

## 6.5 Wave Radar

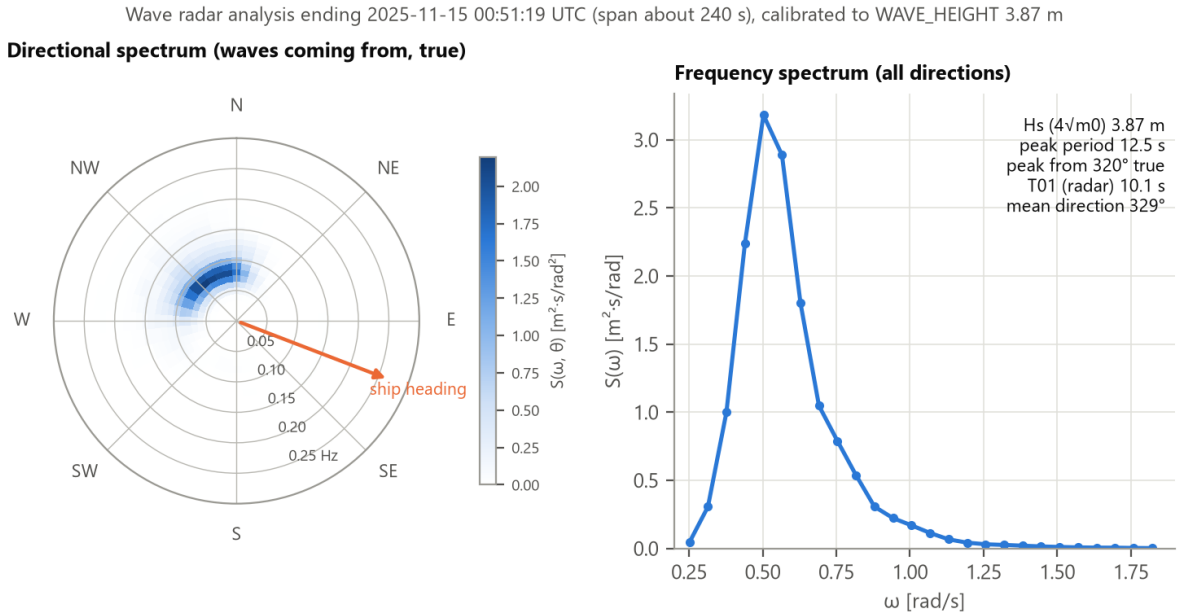



**Figure 12.** Analysis ending 00:51:19 UTC, calibrated (5.4): Hs 3.87 m, peak 12.5 s from ~320° (mean 329°, T01 10.1 s); heading 111°.

### 6.6 Ship Motions

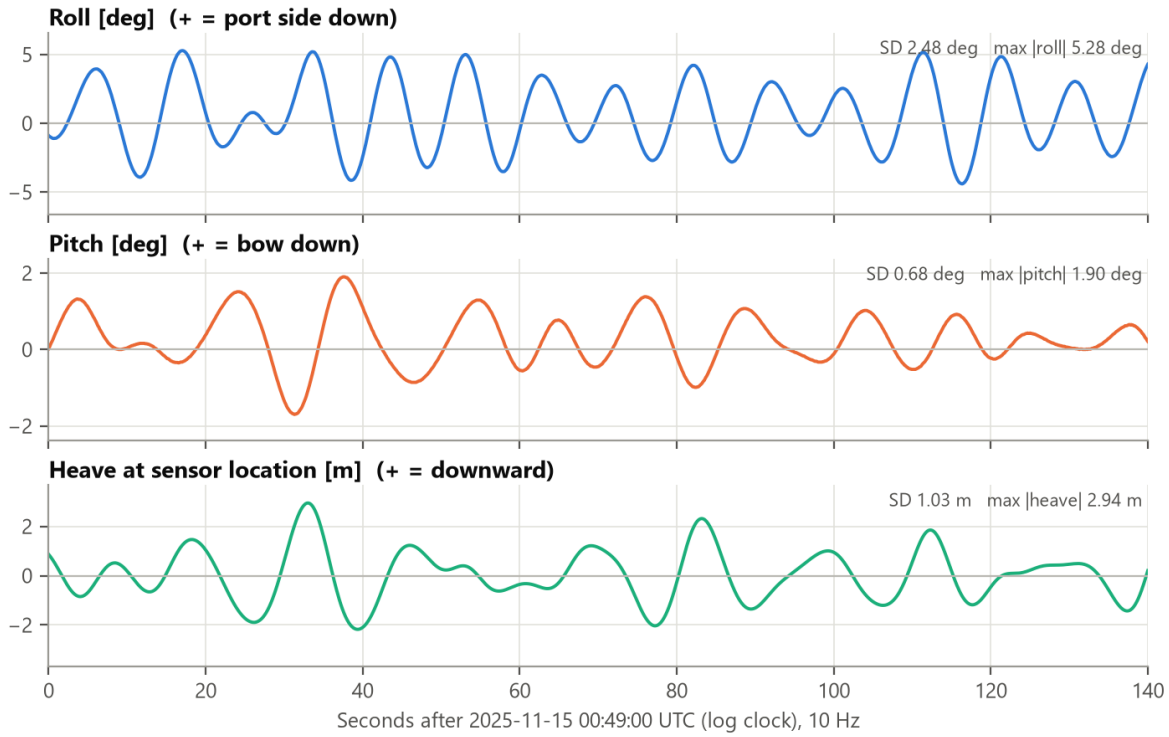



**Figure 13.** Roll, pitch, heave at the IMU, 10 Hz from /history/rtv (signs 5.1). Roll period 9.3–9.7 s (Voy02 leg 2: 9.5 s predicted).

## 7. Notes

- **Purpose:** research and development only
- **Hull:** approximate hull form, identified by file date
- **Mass distribution:** Voy02 leg 2, the condition at the start of the leg; fuel burns off during the leg (~23 t/day)
- **Clocks:** log and radar = DTI receive clock; camera = recorder clock (5.3)
- **Not in the sample:** AIS targets, other days, other report folders
- **Licences:** the DTI player — player\LICENSE.txt (research and development use; no redistribution or reverse engineering); third-party components — player\THIRD-PARTY-NOTICES.txt

Contact: kim@shippartners.jp